University of Copenhagen
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Optimized Virtual Orbital Space for Quantum Computing

Master's Thesis in Chemistry / Quantum Chemistry

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Abstract

⟨*Write the abstract here after all chapters are complete.*⟩

Keywords: Optimized virtual orbital space, OVOS, MP2, orbital optimisation, virtual orbital reduction, quantum computing, PySCF, Newton-Raphson.

Resumé

⟨*Danish translation of the abstract.*⟩

Acknowledgements

⟨*Write acknowledgements here.*⟩

List of Abbreviations

HF	Hartree-Fock
RHF	Restricted Hartree-Fock
UHF	Unrestricted Hartree-Fock
MP2	Second-Order Møller-Plesset Perturbation Theory
MPn	Møller-Plesset Perturbation Theory of order n
CI	Configuration Interaction
CISD	Configuration Interaction Singles and Doubles
FCI	Full Configuration Interaction
CC	Coupled Cluster
CCSD	Coupled Cluster Singles and Doubles
MCSCF	Multi-Configurational Self-Consistent Field
CASSCF	Complete Active Space Self-Consistent Field
DFT	Density Functional Theory
OVOS	Optimized Virtual Orbital Space
FNO	Frozen Natural Orbitals
PNO	Pair Natural Orbitals
DLPNO	Domain-Based Local Pair Natural Orbitals
NO	Natural Orbitals
VQE	Variational Quantum Eigensolver
QPE	Quantum Phase Estimation
JW	Jordan-Wigner
BK	Bravyi-Kitaev
NISQ	Noisy Intermediate-Scale Quantum

QC	Quantum Computing
BO	Born-Oppenheimer
AO	Atomic Orbital
MO	Molecular Orbital
SCF	Self-Consistent Field
LCAO	Linear Combination of Atomic Orbitals
ERI	Electron Repulsion Integral
CBS	Complete Basis Set
GTO	Gaussian-Type Orbital
STO	Slater-Type Orbital
DIIS	Direct Inversion in the Iterative Subspace
NR	Newton-Raphson
RLE	Reduced Linear Equation
PySCF	Python-based Simulations of Chemistry Framework
UCPH	University of Copenhagen

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Chapter 4

Method

4.1 Algorithm Overview

Algorithm 1 OVOS Iterative Optimisation

```
Perform SCF calculation to obtain initial MO coefficients C
Transform AO integrals to MO basis: ⟨ij|ab⟩
Select initial active virtual space
repeat
    Compute MP1 amplitudes t_ij^ab and MP2 energy J_2
    Assemble gradient G and Hessian H
    Solve Newton-Raphson equation: ΔR = −H^{−1} G
    Generate unitary matrix U = exp(R)
    Update MO coefficients: C ← C U
    Canonicalise active Fock block
    Evaluate new J_2
until |J_2^(n) − J_2^(n−1)| < ε_conv
```

4.2 Integral Transformation

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5.1 Software and Libraries

5.2 Hardware

5.3 Molecular Geometries

Table 5.1: Molecular geometries and electronic structure parameters used in this work.

Molecule	Geometry (Å)	Multiplicity	Charge	Symmetry
H_2	$r(\mathrm{H}-\mathrm{H}) = 0.74144$	1	0	$D_{\infty h}$
LiH	$r(\mathrm{Li}-\mathrm{H}) = 1.595$	1	0	$C_{\infty v}$
H_2O	$r = 0.957$, $\angle = 104.5$ř	1	0	C_{2v}
CH_2	planar, see text	3	0	C_{2v}
BH_3	$r(\mathrm{B}-\mathrm{H}) = 1.19$, planar	1	0	D_{3h}
N_2	$r(\mathrm{N}-\mathrm{N}) = 1.10$	1	0	$D_{\infty h}$
CO	$r(\mathrm{C}-\mathrm{O}) = 1.128$	1	0	$C_{\infty v}$
HF	$r(\mathrm{H}-\mathrm{F}) = 0.917$	1	0	$C_{\infty v}$
NH_3	$r(\mathrm{N}-\mathrm{H}) = 1.012$	1	0	C_{3v}

5.4 Basis Sets

5.5 Reference SCF Calculations

5.6 OVOS Calculation Settings

5.7 Data Storage and Reproducibility

Chapter 6

Results

In this chapter, the results of the OVOS implementation are presented. The chapter is structured as follows:

- **Section 6.1** validates the implementation by comparing full-space MP2 correlation energies against PySCF and literature values.
- **Section 6.2** presents the main result: the fraction of MP2 correlation energy recovered as a function of the active virtual orbital count.
- **Section 6.3** analyses the convergence behaviour of the OVOS algorithm.
- **Section 6.4** compares different initialization strategies and their impact on final energies and convergence.
- **Section 6.5** investigates how the choice of basis set affects energy recovery.
- **Section 6.6** provides a direct comparison with the results published by Adamowicz and Bartlett in their original OVOS paper.
- **Section 6.7** assesses the computational cost and scaling of the OVOS algorithm.
- **Section 6.8** examines the T1 amplitudes in the converged OVOS orbitals to test for Brueckner-like properties.
- **Section 6.9** estimates the practical benefits of using OVOS orbitals as a reference state for quantum computing applications, including overlap with FCI states and resource estimates for VQE.

6.1 Validation of the Implementation

PySCF gives me an opportunity to validate my implementation of full space MP2 by comparing with their built-in MP2 module, pyscf.mp.MP2. I present my the deviation $\Delta E_{\text{corr}} = E_{\text{corr}}^{\text{code}} - E_{\text{corr}}^{\text{PySCF}}$ is within numerical precision for all test molecules except CO, which shows a bit larger deviation, in Figure 6.1. As CO's deviation is still within the acceptable range, it confirms that my implementation of the MP2 correlation energy is correct, and that the OVOS code correctly reproduces the full-space MP2 limit when $N'_{\text{act.unocc}} = N_{\text{act.unocc}}$.

Molecule	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	$E_{\text{corr}}^{\text{PySCF}}$ (Hartree)	ΔE_{corr} (Hartree)
CO	-2.12168E-01	-2.12168E-01	2.12862E-08
H_2O	-1.28851E-01	-1.28851E-01	1.91423E-09
HF	-1.28694E-01	-1.28694E-01	-2.47007E-09
NH3	-1.14441E-01	-1.14441E-01	-2.79223E-10

Table 6.1: Validation of the OVOS implementation against PySCF mp.MP2. Deviations ΔE are within numerical precision ($|\delta E| < 10^{-8}$ Hartree). All molecules are run in the 6–31G basis for this validation.

Verifying that the full-space MP2 energies match PySCF's results, any differences observed in the OVOS results are due to the orbital optimization and active space truncation, rather than a bug in the MP2 implementation.

6.2 Energy Recovery as a Function of Active Virtual Orbital Count

The results of the OVOS algorithm for all test molecules in the 6–31G and cc–pVDZ basis sets for start guess RHF, are presented in this section. Explore the correlation energy recovered by OVOS as a function of the number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

CO

...

H2O

...

HF

The OVOS algorithm results for HF in the 6–31G basis through each $N'_{\text{act.unocc}}$ to full space of active unoccupied orbitals are shown in Table 6.2. OVOS recovers increasingly more correlation energy $E_{\text{corr}}^{\text{OVOS}}$ as $N'_{\text{act.unocc}}$ increases, with the percentage of correlation energy recovered reaching 100% at $N'_{\text{act.unocc}} = 6$. We mark the difference the OVOS algorithm makes at each $N'_{\text{act.unocc}}$ from initial MP2 as $\Delta E_{\text{corr}}^{\text{OVOS}}$, highlighted in Figure 6.1 as the points for each $N'_{\text{act.unocc}}$. The most recovered correlation energy is at $N'_{\text{act.unocc}} = 4$ with $\Delta E_{\text{corr}}^{\text{OVOS}} = -5.49038e-2$. Convergence is achieved in all cases, with iteration counts increasing as $N'_{\text{act.unocc}}$ reaches 50% of the total number of active unoccupied orbitals and decreases thereafter.

The table is plotted in Figure 6.1 to show the convergence behaviour of the OVOS algorithm for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

NH3

...

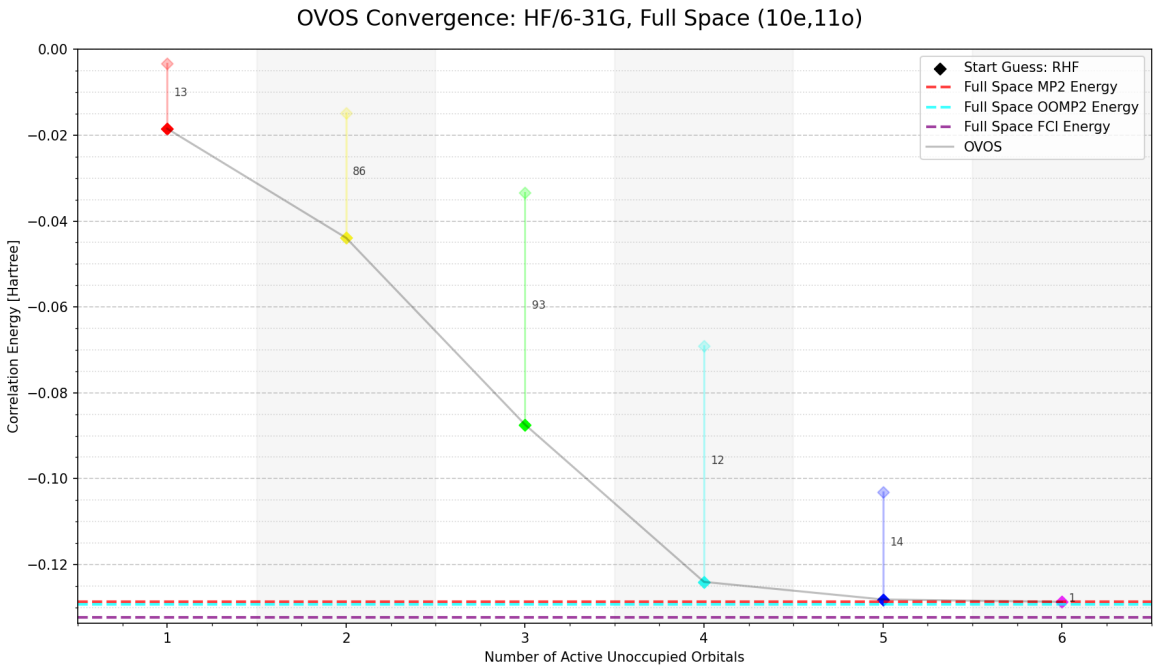

Figure 6.1: Convergence of the MP2 correlation energy for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	-1.84737E-02	14.35%	-1.52027e-2	13
2	-4.39109E-02	34.12%	-2.91136e-2	15
3	-8.74131E-02	67.92%	-5.39991e-2	50
4	-1.24045E-01	96.39%	-5.49038e-2	20
5	-1.28136E-01	99.57%	-2.49738e-2	15
6	-1.28694E-01	100.0%		1

Table 6.2: Summary of OVOS results for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

6.3 Convergence Behaviour

In this section, I present the convergence profiles of the OVOS algorithm for the test molecules in the 6–31G and cc–pVDZ basis sets.

Convergence is achieved when $|\Delta E_{\text{corr}}| < 10^{-8}$ Hartree for at least 50 consecutive iterations. Some cases, would get stuck in a local minimum achieving convergence. The consecutive iteration criterion then avoids these false positives, and help identify true convergence, a global minimum. For example, in the case of HF at $N'_{\text{act.unocc.}} = 3$ in the 6–31G basis, the algorithm converges to a local minimum at iteration 10, but then escapes and continues to converge to the global minimum by iteration 50. This behaviour is illustrated in Figure 6.2.

Placeholder

Figure 6.2: Convergence behavior of the MP2 correlation energy for HF in the 6–31G basis with $N'_{\text{act.unocc}} = 3$ active virtual spin-orbitals. The dashed line marks the convergence threshold of 10^{-8} Hartree.

6.4 Effect of Initialization Strategy

6.5 Basis Set Dependence

6.6 Comparison with Adamowicz and Bartlett (1987)

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Appendix A

Mathematical Derivations

A.1 Spin-Orbital MP2 Energy from the Hylleraas Functional

A.2 Derivation of the Orbital Rotation Gradient

A.3 Derivation of the Orbital Rotation Hessian

A.4 Matrix Exponential for the Rotation Matrix

A.5 Notation Conventions for Two-Electron Integrals

Table A.1: Summary of two-electron integral notation used in this thesis.

Notation	Definition	Used where
$(pq\|rs)$	$\int\int \phi_p^*(1)\phi_q(1)r_{12}^{-1}\phi_r^*(2)\phi_s(2)\,d1\,d2$	PySCF output
$\langle pr\|qs\rangle$	same as $(pq\|rs)$	Physicists' (non-antisym.)
$\langle pq\|\|rs\rangle$	$(pr\|qs) - (ps\|qr)$	Antisymmetrised; used in all formulas

Appendix B

Additional Data and Figures

B.1 Full MP2 Reference Energies

B.2 Complete OVOS Energy Recovery Tables

B.3 Convergence Plots for All Systems

B.4 Hessian Eigenvalue Analysis

Appendix C

Implementation Details and Source Code

C.1 Repository and Version Information

C.2 Spin-Orbital Integral Conversion

Listing C.1: Spatial-to-spin ERI conversion (`spatial_2_spin_eri_optimized`).

```
# Convert spatial ERIs to spin-orbital basis (interleaved alpha/beta)
def spatial_2_spin_eri_optimized(self, eri_aaaa, eri_aabb, eri_bbbb):
    n_spin = self.tot_num_spin_orbs
    eri_spin = np.zeros((n_spin, n_spin, n_spin, n_spin))
    # alpha-alpha block (even indices)
    eri_spin[0::2, 0::2, 0::2, 0::2] = eri_aaaa
    # beta-beta block (odd indices)
    eri_spin[1::2, 1::2, 1::2, 1::2] = eri_bbbb
    # alpha-alpha / beta-beta cross block
    eri_spin[0::2, 0::2, 1::2, 1::2] = eri_aabb
    eri_spin[1::2, 1::2, 0::2, 0::2] = eri_aabb.transpose(2, 3, 0, 1)
    return eri_spin
```

C.3 MP2 Energy and Amplitude Computation

C.4 Orbital Optimisation: Gradient and Hessian

C.5 Profiling Results